$0.9V_{\pi}(s_2) = -1 + 0.8 \times 8 = 5.4$. As you can see, states farther from the winning state are valued less.

Training a reinforcement learning agent, then, just amounts to successfully training a neural network to either approximate the value function (so the agent will choose actions that lead to high-value states) or to directly approximate the policy function by observing rewards after actions and reinforcing actions based on the rewards received. Both approaches have their pros and cons, but often we combine learning both a policy and a value function together, which is called an *actor-critic algorithm*, where the actor refers to the policy and the critic refers to the value function.

11.2. The uncharted topics in deep reinforcement learning

The Markov decision process framework and value and policy functions we just reviewed were detailed in chapters 2–5. We then spent the rest of the book implementing more sophisticated techniques for successfully training value and policy functions in difficult environments (e.g., environments with sparse rewards) and environments with multiple interacting agents. Unfortunately, there were many exciting things we didn't have the space to cover, so we'll end the book with a brief tour of some other areas in deep reinforcement learning you might want to explore. We'll only give a taste of a few topics we think are worth exploring more and hope you will look into these areas more deeply on your own.

11.2.1. Prioritized experience replay

We briefly mentioned the idea of *prioritized replay* earlier in the book when we decided to add multiple copies of the same experience to the replay memory if the experience led to a winning state. Since winning states are rare, and we want our agent to learn from these informative events, we thought that adding multiple copies would ensure that each training epoch would include a few of these winning events. This was a very unsophisticated means of prioritizing experiences in the replay based on how informative they are in training the agent.